Total No. of Questions :3] [2511]-0305161 Seat No.:

Society for Computer Technology and Research's

PUNE INSTITUTE OF COMPUTER TECHNOLOGY

IN-SEM EXAMINATION OCT-2025

(SY) (B.Tech.) (ETE) (SEM-III)

(0305161) (Data Structures and Algorithms)

(2024 PATTERN)

Time : 1 Hours] **[Max Marks : 20**

Instructions to the candidates:

1. *Assume suitable data wherever required.*
2. *Neat diagrams must be drawn wherever necessary.*
3. *Use of non-programmable calculators is allowed.*
4. *[Marks], [Course Outcome] and [Bloom's Taxonomy Level] given at the end of each questions.*
5. ***Attempt Any Two Questions***

Q.1)A) Multiple choice question, each 1 mark [05 M] [CO-1], [L1, L2, L3]

1. Identify the sorting technique in which the smallest element is repeatedly selected and placed at its correct position.

A. Bubble B. Selection
C. Merge D. Quick

2. For a program with global variables and recursion, the memory for program instructions is fixed, while memory used for recursion calls is_____

A. Recursion B. Variable
C. Dynamic D. Iterative

3. The assertion and reasoning are labeled as A and R respectively, choose the correct option

A: Binary search is faster than linear search for large datasets.
R: Binary search eliminates half of the elements in each step.

A. Both A & R true, R explains A

B. Both A & R true, R not explanation

C. A true, R false

D. A false, R true

4. Identify correct statement/s about factorial recursion. Multiple answers may apply.

A. fact (1) = 1 B. fact (0) =1
C. fact(n) = n*fact(n-1) D. fact(n) = fact(n+1)

5. Which of the following problems are suitable for solving using backtracking? (Select all that apply)
A. N-Queens problem B. Sudoku Solver
C. Fibonacci sequence D. Maze solving

B) Differentiate between recursion and backtracking. Explain factorial of a number in detail. [05 M] [CO-01] [L2]

Q.2)A) Compare bubble, insertion, selection, quick, and merge sort algorithms in terms of time complexity, space complexity, stability, in-place operations, recursion, and adaptiveness by organizing the information in a tabular form. [06 M] [CO-01] [L3]

B)Write an executable C++ function to implement binary search on a sorted array to find a given number, and test it with sample input. Assume the array: 18, 28, 38, 48, 58, 68. [04 M] [CO-01] [L3]

Q.3)A) Write a pseudo code that can be directly translated into a program to sort any given array using insertion sort. [04 M] [CO-01] [L2]

B) Develop an executable C++ program to search for the number 10 in the array and output:
1.Total number of occurrences of 10.
2.The positions (indices) where 10 is located.
Test the program with the array {10, 20, 30, 10, 40, 78, 88, 10, 89, 45, 10} and verify the results. [04 M] [CO-01] [L3]

C)Define Space and Time complexity with suitable example. [02 M] [CO-01] [L1]

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